

Anindita Dutta

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Research Profile

CSE graduate of BRAC University, recipient of the **Best Thesis Award** and author of a peer-reviewed **ICCIT 2025** publication. Research and industry experience in **machine learning, predictive modeling, and healthcare data analytics**, spanning data validation, feature engineering, and rigorous model evaluation. Seeking a Master's in **Data Science** to advance data-driven research on intelligent systems.

Education

BRAC University — Bachelor of Science in Computer Science and Engineering 2022 – 2026
CGPA: **3.58/4.00**

Research & Publications

Diffusion Models with Graph Attention for Spatiotemporal EEG Inpainting

Undergraduate Thesis | BRAC University | 2026

- Developed a hybrid deep-learning framework combining **Denoising Diffusion Probabilistic Models (DDPM)**, **Graph Attention Networks (GAT)**, and **Singular Value Decomposition (SVD)** using **DEAP** dataset to reconstruct missing multichannel EEG segments via diffusion-based temporal inpainting.
- Evaluated reconstruction quality using **MSE, RMSE, MAE, Pearson Correlation Coefficient (PCC), and PSD distance**, including ablation experiments.

Temporal Fusion of Convolutional and LSTM Networks for Vision-Based Fall Detection Using Anatomical Keypoints

ICCIT 2025 | Peer-Reviewed Conference Publication | IEEE Xplore

- Developed a fall-detection framework using **YOLOv8 pose estimation** and a **CNN-LSTM** architecture with motion features (velocity, rolling mean, rolling standard deviation), achieving **98.48% accuracy**.

Professional Experience

Data Analyst — Bongo Rithm Healthcare Limited 2026 – Present

- Support a **Chicago-based startup** building an **AI-enabled EMR platform for Bangladesh**: healthcare data **collection, validation, and quality management** with **machine-learning** analysis to improve data quality.

Business Development Intern — BRAC-Aarong Feb 2026 – May 2026

- Conducted **market research, demand analysis, competitor tracking, and data-driven reporting** across inventory, operations, and technical teams.

Technical Skills

Programming & Data: Python, SQL, Pandas, NumPy, MS Excel

ML & AI: PyTorch, DDPM, GAT/GNN, Computer Vision, Feature Engineering

Analysis & Tools: Data Visualization, Statistical Reporting, Tableau, Looker Studio, OpenCV, GitHub

Leadership & Academic Activities

Director of Finance — BRAC University Computer Club 2024 – 2026

- Managed budgets of up to BDT 400,000 for a 500+ member organization: financial planning, budgeting, and operations for large-scale technical events.

Member — Machine Vision & AI Subteam, BRACU Duburi 2024 – 2025

- Contributed to computer vision projects using **OpenCV & Docker**: dataset labeling, image preprocessing, and technical support for model development.

Awards

Best Thesis Award — BRAC University 2026

Recognized for outstanding undergraduate research in deep generative modeling

Presidential Award — National Award Winner for Singing 2007